JABchem

Past Papers Advanced Higher Chemistry

2025

Marking Scheme

Grade Awarded	Mark Required (/160)	%	% candidates achieving grade
A	108+	68%	30.0%
B	92+	58%	25.0%
C	76+	48%	22.2%
D	60+	38%	13.8%
No award	<60	<38%	9.0%

Section:	Multiple Choice	Extended Answer	Project
Average Mark:	17.5 /25	49.1 /95	27.1 /40

2025 Adv Higher Chemistry Marking Scheme

<table>
<tr><th>MC Qu</th><th>Answer</th><th>% Correct</th><th>Reasoning</th></tr>
<tr><td>1</td><td>A</td><td>93</td><td>☑A each line corresponds to the exact amount of energy released when an electron drops down from a higher energy level to a lower energy level.
☒B visible part of electromagnetic spectrum is from 380nm to 750nm
☒C energy is absorber not emitted when electron moves to higher energy level
☒D wavelengths emitted are from energy released when electrons drop down energy levels</td></tr>
<tr><td>2</td><td>D</td><td>67</td><td>☒A all p-orbitals have a figure of 8 shape but no ring in middle as in the diagram
☒B all p-orbitals have a figure of 8 shape but no ring in middle as in the diagram
☒C only d-orbital dz^2 has the shape shown, the other four have double figure of 8 shapes
☑D the dz^2 orbital has the shape shown in the diagram</td></tr>
<tr><td>3</td><td>D</td><td>61</td><td>X^{3+} ion has 55 electrons ∴ an atom of X must have 58 electrons and therefore 58 protons
Element with atomic number of 58 is Cerium Ce and is one of the Lanthanides in f-block</td></tr>
<tr><td>4</td><td>B</td><td>53</td><td>$$\text{Electron Pairs} = \frac{\text{No. of electrons in outer shell of central atom} + \text{No. of bonds} - \text{Charge}}{2}$$
$$\text{Electron Pairs} = \frac{6 + 4 - 0}{2}$$
$$\text{Electron Pairs} = \frac{10}{2} = 5 \text{ pairs}$$ = Trigonal bipyramidal electron pair arrangement</td></tr>
<tr><td>5</td><td>A</td><td>42</td><td><table>
<tr><td>Complex</td><td>$[Cu(OH_2)_6]^{2+}$</td><td>$[CuCl_4]^{2-}$</td></tr>
<tr><td>Light Transmitted</td><td>blue</td><td>green</td></tr>
<tr><td>Light Absorbed</td><td>yellow</td><td>purple</td></tr>
<tr><td>Wavelengths Absorbed</td><td>590nm-595nm</td><td>380nm-400nm</td></tr>
<tr><td>Energy of light absorbed</td><td>Higher wavelength = Lower energy</td><td>Lower wavelength = higher energy</td></tr>
</table>
☑A increase in energy of light absorbed and increase in splitting of d-orbitals as a result
☒B increases the splitting of d-orbitals increases the energy of the light absorbed
☒C energy of light absorbed increases
☒D energy of light absorbed increases</td></tr>
<tr><td>6</td><td>C</td><td>39</td><td>☒A all 5 d-orbitals are only degenerate (equal in energy) when there are no ligands present
☒B the arrangement is the effect of octahedral complex not a tetrahedral arrangement
☑C in octahedral arrangement dxy, dxz and dyz are lower in energy as they lie along the x,y and z axes while the dx^2-y^2 and dz^2 orbitals are higher in energy because there are not aligned with the x, y and z axes and repelled more by the ligands
In the tetrahedral arrangement the situation is reversed. The dxy, dxz and dyz do not lie along the axes of the tetrahedral arrangement are they are pushed higher in energy by repulsion from ligands and the dx^2-y^2 and dz^2 orbitals are repelled by the ligands less
☒D the arrangement is named after legendary Italian international fullback Fuctivano</td></tr>
<tr><td>7</td><td>A</td><td>48</td><td>☑A Elemental sulphur has oxidation number zero and S in $S_2O_3^{2-}$ ion is +2
☒B if 3O=-6 and charge of ion is -2 then 2S= +4 ∴ S in $S_2O_3^{2-}$ has oxidation number +2
☒C S_8 is the element sulphur. All elements have an oxidation number of zero
☒D S_8 is the element sulphur. All elements have an oxidation number of zero</td></tr>
<tr><td>8</td><td>C</td><td>90</td><td><table>
<tr><th>Brønsted-Lowry Term</th><th>Definition</th></tr>
<tr><td>Acid</td><td>Proton donor to form conjugate base</td></tr>
<tr><td>Base</td><td>Proton acceptor to form conjugate acid</td></tr>
<tr><td>Conjugate Acid</td><td>Formed when Base accepts proton</td></tr>
<tr><td>Conjugate Base</td><td>Formed when Acid loses proton</td></tr>
</table></td></tr>
</table>

9 | D | 89

$$pH = pK_a - \log_{10}\frac{[Acid]}{[Salt]}$$

$$pH = 4.83 - \log_{10}\frac{0.100}{0.200}$$

$$pH = 4.83 - \log_{10}(0.5)$$

$$pH = 4.83 - (-0.30)$$

$$pH = 5.13$$

10 | A | 47

☑A NaCl is the salt formed by reaction of strong acid and strong alkali ∴ pH of salt = 7
☒B Na_2CO_3 is the salt formed by reaction of weak acid and strong alkali ∴ pH of salt > 7
☒C Na_2SO_3 is the salt formed by reaction of weak acid and strong alkali ∴ pH of salt > 7
☒D CH_3COONa is the salt formed by reaction of weak acid and strong alkali ∴ pH of salt > 7

11 | B | 71

Property	0.1 mol l^{-1} ethanoic acid	0.1 mol l^{-1} hydrochloric acid
pH	Higher (partial dissociation = fewer H^+ ions)	Lower (full dissociation = more H^+ ions)
Conductivity	Lower (partial dissociation = fewer ions to conduct)	Higher (full dissociation = more ions to conduct)
Rate of Reaction with Magnesium	Slower (partial dissociation = fewer H^+ ions to react)	Faster (full dissociation = more H^+ ions to react)

12 | B | 80

Number of each reactant in the rate determining step givers the order for each reactant.

- 2 molecules of NO in r.d.s. ∴ order of reaction with respect to NO = 2
- 1 molecule of H_2 in r.d.s. ∴ order of reaction with respect to H_2 = 1

Rate equation for reaction rate = $k[NO]^2 [H_2]$ and overall order = 2+1 = 3

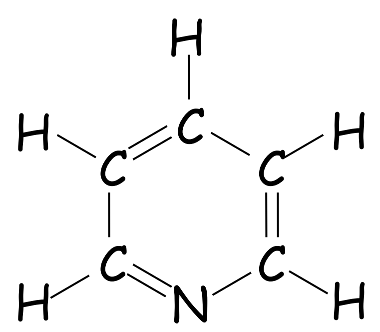
13 | D | 79

The energy change for the formation of one mole of a substance from elements in their standard state
☒A oxygen needs to be O_2 molecules for the element to be in its standard state
☒B none of the three reactants are in their standard state
☒C calcium is a solid in its standard state
☑D reactants are in their standard states and 1 mole of the substance if formed

14 | C | 76

A reaction with a positive S° value is a reaction with an increase in disorder
☒A Reaction has a decrease in disorder as two reactants join to become one product
☒B Reaction has a decrease in disorder as two reactants join to become one liquid product
☑C Reaction has an increase in disorder as a solid is removed and a gas is formed
☒D Reaction has a decrease in disorder as a gas is removed and solid is formed

15 | C | 66



Type of Bond	C - C	C = C	C - H	N=C	N-C
Number of bonds	2	2	5	1	1

Type of Bond		C - C	C = C	C - H	N=C	N-C
Sigma	σ	1	1	1	1	1
Pi	π	0	1	0	1	0

Sigma bonds = 2 C - C + 2 C = C + 5 C - H + 1 N=C + 1 N-C
= 2σ + 2σ + 5σ + 1σ + 1σ
= 11σ

16 | B | 74

☒A Both can be made by nucleophilic substitution
☑B pure ethers like ethoxyethane do not have hydrogen bonding between molecules
☒C both compounds have the formula $C_4H_{10}O$ and have the same gram formula mass
☒D both compounds are carbon-based and are flammable

17 | A | 48

Answer	Type of Compound: Compound 1	Type of Compound: Compound 2
A	tertiary haloalkane	tertiary amine
B	primary amine	tertiary haloalkane
C	primary haloalkane	tertiary amine
D	secondary amine	secondary haloalkane

18	C	70	☒A addition reactions do not take place on benzene rings ☒B nitrate NO_3^- ions are not involved in this electrophilic substitution reaction ☑C NO_2^+ is the electrophile in this electrophilic substitution reaction ☒D electrophiles are attracted to the delocalised electrons in benzene.
19	D	76	☒A optical isomers (enantiomers) are mirror images of each other ☒B geometric isomers can be based around C=C double bonds and ring structures ☒C there can be differences in physical & chemical properties due to the difference shapes ☑D C=C double bonds and ring structures can provide the restricted rotation needed for geometric isomerism where two different groups are attached across C=C or ring
20	C	48	☒A n= v x c = 0.2litres x0.12mol l^{-1} = 0.024mol f.u. but 3 Cl^- ions per f.u. ∴ 0.072mol -ve ions ☒B n= v x c = 0.3litres x0.15mol l^{-1} = 0.045mol f.u. but 1 I^- ions per f.u. ∴ 0.045mol -ve ions ☑C n= v x c = 0.4litres x0.10mol l^{-1} = 0.040mol f.u. but 2 NO_3^- ions per f.u. ∴ **0.080mol -ve ions** ☒D n= v x c = 0.5litres x0.10mol l^{-1} = 0.050mol f.u. but 1 SO_4^{2-} ions per f.u. ∴ 0.050mol -ve ions
21	A	74	☑A radio waves are used to flip hydrogen nuclei from lower to higher energy alignment ☒B ultraviolet is used in the initiation reaction in free radical chain reactions ☒C Infrared is absorbed by specific bonds allowing the identification of type of bond ☒D visible light is absorbed in d-d transitions is transition metals transmitting another colour
22	B	54	☒A sodium hydroxide cannot be used as a primary standard as it absorbs moisture from air ☑B sodium carbonate is a primary standard used to standardise acid like hydrochloric acid ☒C potassium dichromate needs acid to become a oxidising agent.Not used to standardise acid ☒D potassium hydrogen phthalate is used to standardise alkalis not acids
23	D	74	☒A impurities lower melting point ∴ purification should increase melting point ☒B impurities lower melting point ∴ purification should increase melting point ☒C impurities widen the melting point ∴ purification would narrow the range ☑D purification should increase the melting point and narrow the range
24	B	42	☒A carotene is a hydrocarbon so water cannot be used as the solvent for carotene ☑B carotene is a hydrocarbon and is soluble in hexane and green-blue filter should be used ☒C green-blue filter is used as it is complimentary to the colour orange ☒D green-blue filter is used as it is complimentary to the colour orange
25	C	55	☒A Step 1 does not remove soluble impurities as they are dissolved into the hot solvent ☒B Hot filtration of resulting mixture in Step 2 to remove insoluble impurities ☑C Hot filtration removes insoluble impurities. Filtering, washing and drying separates soluble impurities ☒D Hot filtration of resulting mixture in Step 2 to remove insoluble impurities

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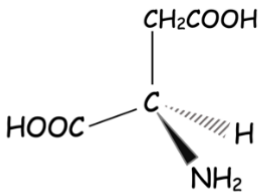
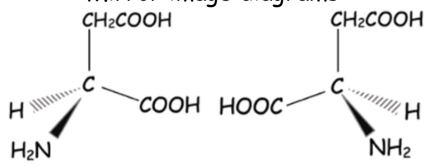
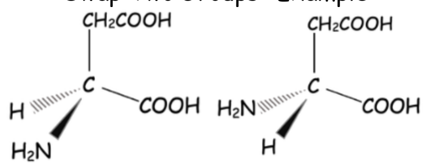
Long Qu	Answer	Reasoning
1a	non-superimposable mirror images	Optical isomers are asymmetric non-superimposable mirror images formed when a chiral carbon has four different groups attached to it.
1b		Mirror image diagrams  L-aspartic Acid D-aspartic Acid or "Swap Two Groups" Example  L-aspartic Acid D-aspartic Acid
1c	One answer from:	rotates by same {amount / angle} in {opposite / different} directions
1d	racemic	A racemic mixture has equal concentrations of both optical isomers and this makes the solution optically inactive giving no rotation of plane-polarised light.
2a	Answer to include:	Side-on overlap of unhybridised p-orbitals (perpendicular to the σ-axis)
2b(i)	One answer from:	the more hydroxyl/hydroxy/OH groups the stronger the acid the longer the alkyl group/carbon chain the stronger the acid ethyl phenols are stronger than methyl phenols if the alkyl/methyl/ethyl/carbon chain group is on carbon number 4 position the acid is stronger (than if it is on carbon number 2 position) the presence of an alkyl/methyl/ethyl group lowers the strength.
2b(ii)	2.77×10^{-6}	$pH = \frac{1}{2}pK_a - \frac{1}{2}\log_{10}c$ $pH = \frac{1}{2}\times(10.29) - \frac{1}{2}\log_{10}(0.150)$ $pH = 5.145 - \frac{1}{2}\times(-0.824)$ $pH = 5.145 - (-0.412)$ $pH = 5.56$ $pH = 5.56$ $-\log_{10}[H^+] = 5.56$ $\log_{10}[H^+] = -5.56$ $[H^+] = 10^{-5.56}$ $[H^+] = 2.77\times10^{-6}$ mol l^{-1}
2c(i)	Answer to include:	1st mark (one from:) absorbs {higher energy / shorter wavelength / higher frequency} changes from absorbing yellow-green to green-blue or 560-580 nm to 480-490 nm larger energy gap between HOMO and LUMO 2nd mark (one from:) fewer atoms/electrons/bonds in the conjugated system shorter/smaller conjugated system conjugation decreases
2c(ii)	4.1-6.1 or 5.1±1	$pH = pK_{In} \pm 1$ $pK_{In} = -\log_{10}(7.9\times10^{-6})$ 5.1 pH range = 5.1 ± 1 $= 4.1\text{-}6.1$
3a	gravimetric	Gravimetric analysis is used to determine the mass of an element or compound in a substance. The substance is converted into another substance which can be readily isolated and purified. In this example, the chloride ions in seas water are removed by a precipitation reaction using silver (I) nitrate and collected by filtration and mass measured.
3b	Add more silver(I) nitrate solution (to filtrate to see if there is any more precipitation)	Once the precipitate has been removed by filtration then the filtrate can be tests with a small addition of silver(I) nitrate solution to closely observe if there is further precipitation. • Any precipitation observed means that not all Cl^- ions have been turned reacted • No further precipitation means that all Cl^- ions have been precipitated out.

<table>
<tr><td>3c</td><td>19.3</td><td>

gfm $AgCl$ = (1x107.9)+(1x35.5) =. 107.9 + 35.5 = 143.4g

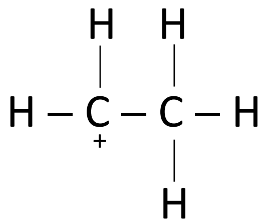
no. of mol = $\frac{\textbf{mass}}{\textbf{gfm}}$ = $\frac{0.779}{143.4}$ = 5.43×10^{-3} mol $AgCl$ = 5.43×10^{-3} mol Cl^-

$20cm^3$ diluted seawater $\longleftrightarrow$ 5.43×10^{-3} mol
$100\ cm^3$ diluted seawater $\longleftrightarrow$ 5.43×10^{-3} mol x $^{100}/_{20}$ = 0.0271mol

$\therefore$ $50cm^3$ seawater $\longleftrightarrow$ 0.0271mol
$1000cm^3$ seawater $\longleftrightarrow$ 0.0271mol x $^{1000}/_{50}$ = 0.543 mol

mass = **n**o. of mol x **gfm** = 0.543 x 35.5 = 19.3 g per litre

</td></tr>
<tr><td>4a(i)A</td><td>



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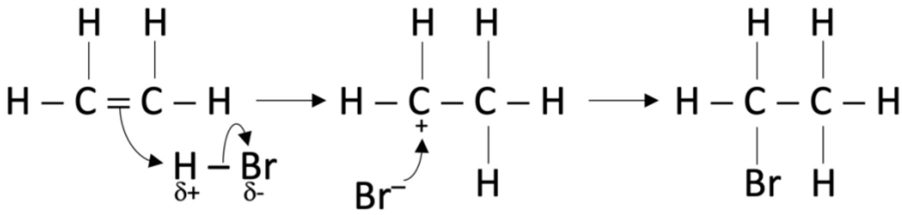
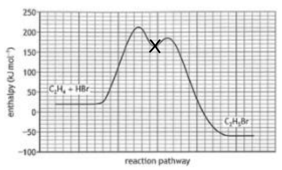


</td></tr>
<tr><td>4a(i)B</td><td>



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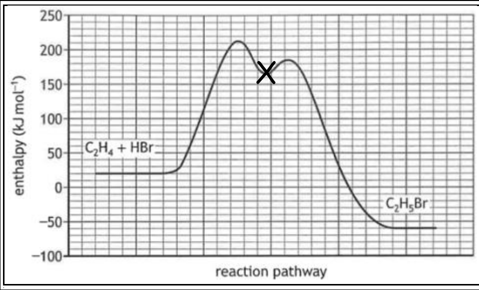


When the intermediate forms, the temporary increase in stability. This stability lowers the enthalpy a little and hence the dip at the top of the hill where the intermediate is formed.

</td></tr>
<tr><td>4a(ii)A</td><td>One answer from:</td><td>

<table><tr><td>entropy of surroundings is increasing</td><td>total entropy (of system and surroundings) is increasing</td><td>reaction is exothermic or ΔH is negative</td><td>heat energy is released</td></tr></table>

</td></tr>
<tr><td>4a(ii)B</td><td>-80</td><td>$\Delta H = H_{products} - H_{reactants} = -60 - 20 = -80\ kJ\ mol^{-1}$</td></tr>
<tr><td>4a(ii)C</td><td>606 or 610</td><td>$\Delta G^o = \Delta H^o - T\Delta S^o = 0 \quad \therefore\ T = \frac{\Delta H^o}{\Delta S^o} = \frac{-80 \times 1000\ J\ mol^{-1}}{-132\ J\ K^{-1}\ mol^{-1}} = 606K$</td></tr>
<tr><td>4b(i)</td><td>

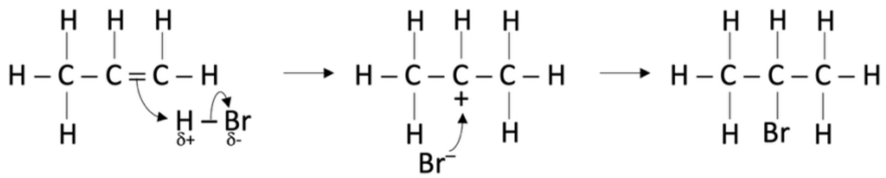
more stable/stabilisation

or

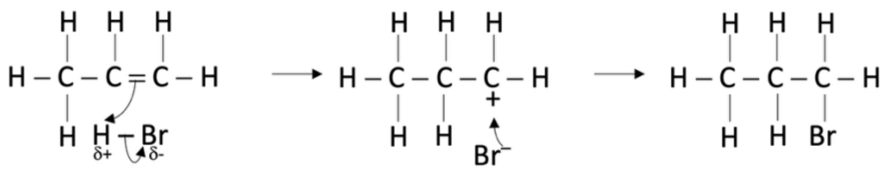
the intermediate for the minor product is less stable

</td><td>

H from H-Br more likely to attach to end carbon as there is more space for the H-Br molecule to collide with the end carbon. This results in 2-bromopropane being the major product in accordance with Markovnikov's Rule.



H from H-Br less likely to attach to middle carbon as there is less space for the H-Br molecule to collide with the middle carbon. This results in 1-bromopropane being the minor product in accordance with Markovnikov's Rule.



</td></tr>
<tr><td>4b(ii)</td><td>

Open Ended Question:

<table><tr><td colspan="4">Marks Awarded</td></tr><tr><td>3 marks</td><td>2 marks</td><td>1 mark</td><td>0 marks</td></tr><tr><td>3%</td><td>13%</td><td>35%</td><td>49%</td></tr></table>

</td><td>

<table><tr><th>3 mark answer</th><th>2 mark answer</th><th>1 mark answer</th></tr><tr><td>Demonstrates a <u>good</u> understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a <u>reasonable</u> understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a <u>limited</u> understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>

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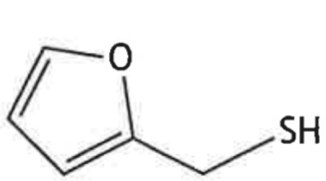
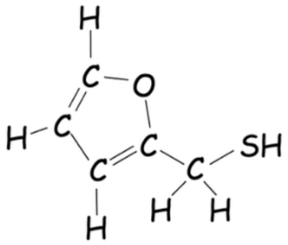
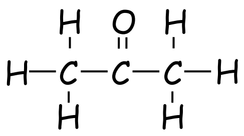
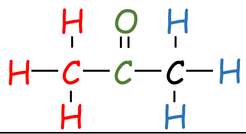
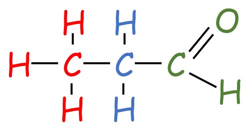
</table>

5a(i)	C_5H_6OS	 Skeletal structure  Full Structural Formula; C_5H_6OS (any order of elements) Molecular Formula
5a(ii)	$[SH]^+$ or ^+SH	The thiol group is a sulfur bonded to a hydrogen atom. Sulphur has r.a.m.=32 and hydrogen has r.a.m.=1 ∴ $[SH]^+$ fragment has m/z = 33
5b(i)	C=O or carbonyl	The infrared adsorption at 1710cm^{-1} is due to the stretching of an C=O bond in an aldehyde or ketone. (The only other infrared stretch at 1710cm^{-1} is the C=O bond in a carboxylic acid but it can be ruled out because carboxylic acids contain 2 oxygen atoms per molecule but the formula of the molecule is C_3H_6O)
5b(ii)A		There is only one peak in the Proton NMR spectrum at chemical shift = 2.2. This corresponds to aldehyde or ketone $CH_3C=O$.
5b(ii)B	3	see below

Propanone (Compound A)	Propanal (Compound B)
	
There should be 1 vertical line for propanone corresponding to	There should be 3 vertical lines for propanal corresponding to

CH_3-C=O	O=C-CH_3
2.7-2.0	2.7-2.0

There is only one vertical line as the two side are identical.

CH_3-R	R-CH_2-CHO	-CH_2-CHO
1.5-0.9		2.7-2.0

5c(i)

Element	Mass in Compound
Carbon	0.401g
Hydrogen	0.067g
Oxygen	**0.532g**

mass of C in compound	mass of H in compound	Mass of O in compound
mass C = $\frac{12}{44}$ x 1.47 = 0.401g	mass C = $\frac{2}{18}$ x 0.60 = 0.067g	= 1.00 - 0.401 - 0.067 = 0.532g

5c(ii) CH_2O

Element	C	H	O
%	0.401	0.067	0.532
No. of moles (divide % by gfm)	$\frac{0.401}{12}$ = 0.033	$\frac{0.067}{1}$ = 0.067	$\frac{0.532}{16}$ = 0.033
Mole ratio (divide through by smallest value)	$\frac{0.033}{0.033}$ = 1.00	$\frac{0.067}{0.033}$ 2.03	$\frac{0.033}{0.033}$ 1.00
Round to Whole Number	1	2	1

5d(i) Answer to include:

1st Mark	separating funnel or a clearly drawn diagram		
2nd Mark (all three required)	shake/mix	leave to separate (into layers)	run off (lower) layer

5d(i) Answer to include:

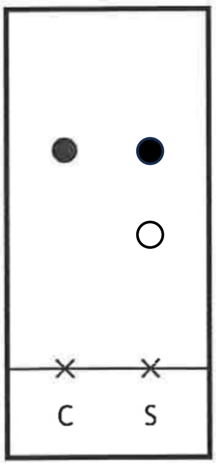
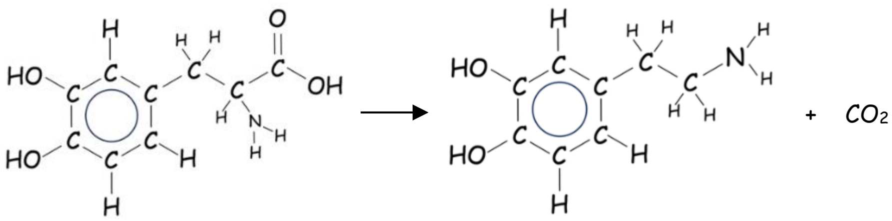
All 4 steps	2 or 3 steps	Only 1 step
2 marks	1 mark	0 marks

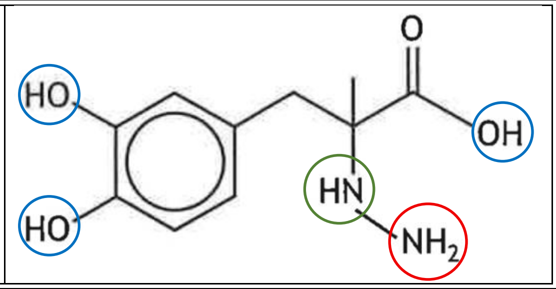
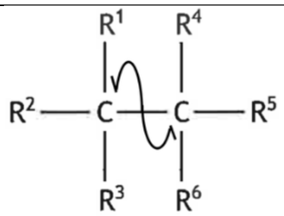
shake/mix/invert in a separating funnel	leave to separate/settle (into layers)	drain/remove/run off (lower) layer	repeat (extraction with fresh dichloromethane) or add more dichloromethane

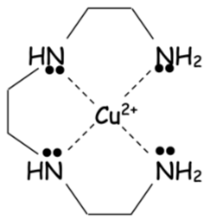
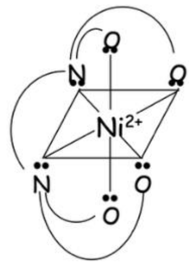
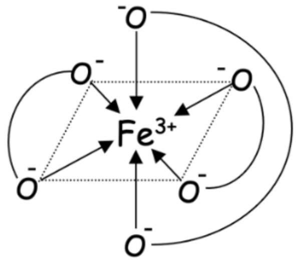
5d(ii) 9.71

Mass of caffeine in aqueous layer = 0.150 - 0.136 = 0.014g

$$K = \frac{[\text{caffeine}]_{\text{dichloromethane}}}{[\text{caffeine}]_{\text{water}}} = \frac{0.136}{0.014} = 9.71$$

5d(iii)		● The added black dot represents the caffeine in the impure sample (S) <u>On Diagram</u>: The must be a dot to represent the caffeine present at the same distance travelled by the pure caffeine (C) ○ The white dot represents impurities. The number of impurity dots will depend on the number of impurities in the sample. <u>On Diagram</u>: The distance travelled by the impurity dot(s) depends on the polarity and size of the impurities. At least one dot must be drawn either vertically above or below the caffeine dot. Multiple dots are also correct
6a(i)	1 mark for: $25cm^3$ or 0.025 litres 1 mark for: use of pipette/burette <u>and</u> making up to the mark/line (with deionised water)	**n**o. of mol = **v**olume x **c**oncentration = 0.5 litres x 0.01 $mol\ l^{-1}$ = 0.005 mol $\text{volume} = \dfrac{\text{no. of mol}}{\text{concentration}} = \dfrac{0.005\ mol}{0.2\ mol\ l^{-1}} = 0.025\ \text{litres} = 25cm^3$ $25cm^3$ of 0.2 mol l^{-1} $KMnO_4$ solution pipetted into $500cm^3$ volumetric flask → Flask filled up to near mark with deionised water and a dropper used to add few drops up to mark on flask. → Mix thoroughly before use.
6a(ii)	65.8	**n**o. of mol = **v**olume x **c**oncentration = 0.0118 litres x 0.010 $mol\ l^{-1}$ = 0.000118 mol $MnO_4^- + 8H^+ + 5Fe^{2+} \longrightarrow Mn^{2+} + 4H_2O + 5Fe^{3+}$ 1mol 5mol 0.000118 mol 0.00059 mol **mass** = **n**o. of mol x **gfm** = 0.00059 x 55.8 = 0.0329g $25cm^3$ sample of solution contains 0.0329g $250cm^3$ flask contains 0.0329g x $^{250}/_{25}$ = 0.329g 5 tablets contains 0.329g 1 tablet contains 0.329g x $^1/_5$ = 0.0658g = 65.8mg
6a(iii)	acidified permanganate would oxidise Cl^- ions to Cl_2	Acidified permanganate solution is a strong enough oxidising agent to cause the oxidation of Cl^- ions into Cl_2 gas $2MnO_4^- + 16H^+ + 10Cl^- \longrightarrow 2Mn^{2+} + 8H_2O + 5Cl_2$
6b(i)	Answer to include:	1st mark: purple / 380-400nm / complementary colour to green } is absorbed 2nd mark: Electrons move to higher energy d orbitals
6b(ii)	One answer from both sections required for 1 mark:	one answer from: the water ligand / water in complex / OH_2 } { donates a proton/H^+ / acts as an acid <u>and</u> one answer from: the water molecule / water solvent / H_2O } { accepts a proton/H^+ / acts as a base
7a	CO_2 or carbon dioxide	 $C_9H_{11}O_4N \longrightarrow C_8H_{11}O_2N + CO_2$

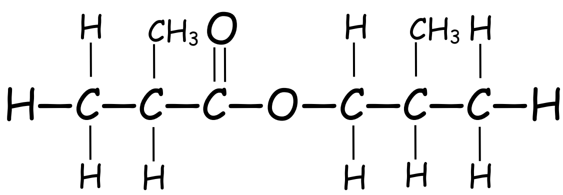
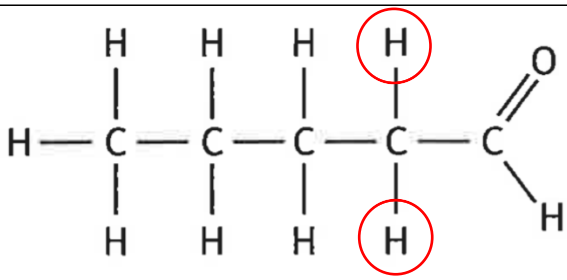
<table>
<tr><td>7b(i)</td><td>1.03×10^{-4}
or
0.000103</td><td>mass = 125mg = 0.125g   gfm = 197g
$\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{0.125}{197} = 6.35\times10^{-4}\text{ mol}$
100% Yield = 6.35×10^{-4} mol
70.5% Yield = 6.35×10^{-4} mol x $^{70.5}/_{100}$
= 4.47×10^{-4} mol
$\text{concentration} = \frac{\text{no. of mol}}{\text{volume}} = \frac{4.47\times10^{-4}\text{ mol}}{4.35\text{ litres}} = 1.03\times10^{-4}\text{ mol l}^{-1}$</td></tr>
<tr><td>7b(ii)A</td><td>(enzyme) inhibitor</td><td><table>
<tr><th>Classification</th><th>Definition</th></tr>
<tr><td>Agonist</td><td>Mimics natural compound and binds to receptors to produce response</td></tr>
<tr><td>Antagonist</td><td>Prevents the natural compound from binding to receptor blocking response</td></tr>
<tr><td>Inhibitor</td><td>Binds to enzyme active site preventing the. reaction catalysed by enzyme</td></tr>
</table></td></tr>
<tr><td>7b(ii)B</td><td>6</td><td>Three groups in compounds can exhibit hydrogen bonding
• 3 from 3x-OH
• 1 from -NH-
• 2 from $-NH_2$

</td></tr>
<tr><td>8a</td><td>254</td><td>$E = \frac{L \times h \times c}{\lambda} = \frac{6.02\times10^{23}\text{ mol}^{-1} \times 6.63\times10^{-34}\text{ J s} \times 3\times10^{8}\text{ m s}^{-1}}{471\times 10^{-9}\text{m}}$
$= 254220\text{ J mol}^{-1}$
$= 254\text{ kJ mol}^{-1}$</td></tr>
<tr><td>8b(i)</td><td>
</td><td>Homolytic fission happens when a pair of electrons split evenly and one electron goes to each end of the bond creating two free radicals.
Curly arrows with a single head (shown here) represent the movement on one electron only.</td></tr>
<tr><td>8b(ii)</td><td>One answer from:</td><td><table><tr><td>(Complex) mixture of products (formed)</td><td>Many products (formed)</td><td>Side reactions</td></tr></table></td></tr>
<tr><td>9</td><td>Open Ended Question:<table>
<tr><th colspan="4">Marks Awarded</th></tr>
<tr><td>3 marks</td><td>2 marks</td><td>1 mark</td><td>0 marks</td></tr>
<tr><td>10%</td><td>32%</td><td>40%</td><td>18%</td></tr>
</table></td><td><table>
<tr><th>3 mark answer</th><th>2 mark answer</th><th>1 mark answer</th></tr>
<tr><td>Demonstrates a <u>good</u> understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a <u>reasonable</u> understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a <u>limited</u> understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr>
</table></td></tr>
<tr><td>10a(i)</td><td>904cm^3</td><td>mass = 1.00g   gfm = 197g
$\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{1.00}{197} = 0.00508\text{ mol}$
$2Au + 4NaCN + \frac{1}{2}O_2 + H_2O + 2OH^- \longrightarrow 2[Au(CN)_2]^- + 4NaOH$
2mol   4mol
0.00508mol   0.0102mol
gfm NaCN. = (1x23)+(1x12)+(1x14) = 23+12+14 = 49g
mass = no. of mol x gfm = 0.0102 x 49 = 0.497g
0.0550%. = 0.00550g per 100cm^3
0.005500g in 100cm^3
0.497g in $100\text{cm}^3 \times {}^{0.497}/_{00.00550} = 904\text{cm}^3$</td></tr>
</table>

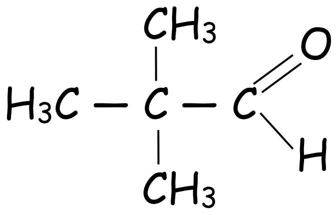
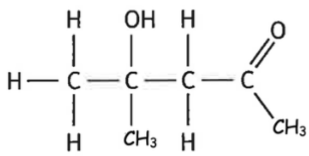
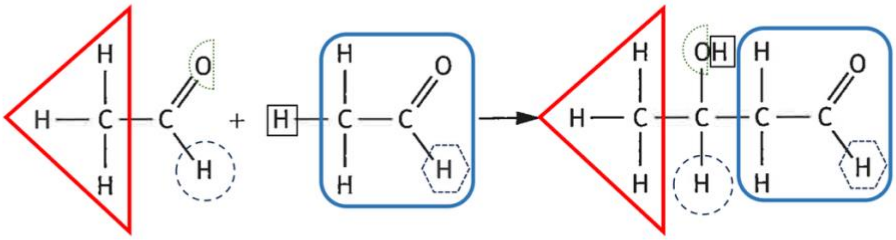
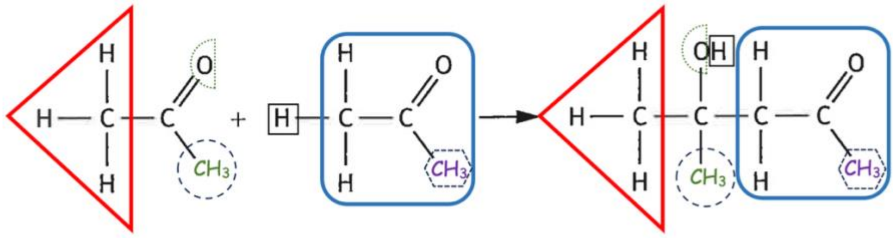
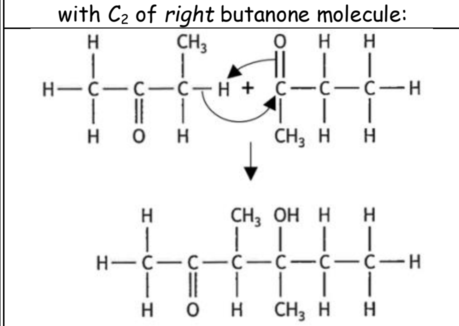
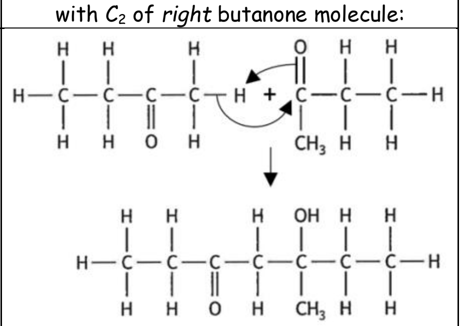
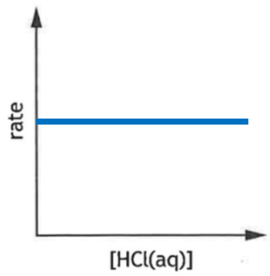
<table>
<tr>
<td>10a(ii)A</td>
<td>total number of bonds from ligands to central metal ion
or
number of bonds the metal ion forms
or
number of pairs of electrons donated/shared to the metal ion</td>
<td>Examples of ligands forming complexes with central transition metal ions:
<table>
<tr>
<td>
</td>
<td>
</td>
<td>
</td>
</tr>
<tr>
<td>coordination number = 4</td>
<td>coordination number = 6</td>
<td>coordination number = 6</td>
</tr>
</table>
</td>
</tr>
<tr>
<td>10a(ii)B</td>
<td>dicyanidoaurate(I)</td>
<td>

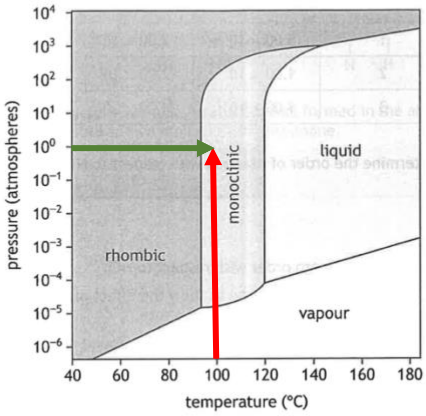
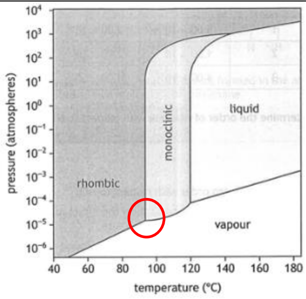
$\underbrace{\text{di}}_{\text{no. of ligands}}\underbrace{\text{cyanido}}_{\text{CN}^- \text{ ligand}}\underbrace{\text{aur}}_{\text{metal name}}\underbrace{\text{ate}}_{\text{negative complex}}\underbrace{\text{(I)}}_{\text{charge on metal ion}}$ = $[Au(CN)_2]^-$

<table>
<tr>
<td>Neutral ligands include:</td>
<td>Negative Ligands include:</td>
<td>Central Ion:</td>
<td>Charge:</td>
</tr>
<tr>
<td>
<table>
<tr><td>Ligand</td><td>Name</td></tr>
<tr><td>OH_2</td><td>aqua</td></tr>
<tr><td>NH_3</td><td>ammine</td></tr>
<tr><td>CO</td><td>carbonyl</td></tr>
</table>
</td>
<td>
<table>
<tr><td>Ligand</td><td>Name</td></tr>
<tr><td>Chloride Cl^-</td><td>chlorido</td></tr>
<tr><td>Cyanide CN^-</td><td>cyanido</td></tr>
<tr><td>Nitrite NO_2^-</td><td>nitrito</td></tr>
</table>
</td>
<td><u>Positive Complex</u>:
metals keep their name
<u>Negative Complex</u>:
Metals end in ATE
e.g. Cuprate, Ferrate, Cobaltate</td>
<td>Charge of central ion is converted into roman numerals and put in brackets</td>
</tr>
</table>
NB: Multiple ligands are listed in alphabetical order ∴ ammine listed before chloride
</td>
</tr>
<tr>
<td>10b(i)</td>
<td>reduction</td>
<td>Reduction: $Au^+ + e^- \longrightarrow Au$</td>
</tr>
<tr>
<td>10b(ii)</td>
<td>Answer to include:
<table>
<tr><td>All 4 steps</td><td>2 or 3 steps</td><td>Only 1 step</td></tr>
<tr><td>2 marks</td><td>1 mark</td><td>0 marks</td></tr>
</table>
</td>
<td>
<table>
<tr><td>filter</td><td>wash</td><td>dry/heat</td><td>to constant mass</td></tr>
</table>
</td>
</tr>
<tr>
<td>10b(iii)</td>
<td>50.8</td>
<td>21.5ppm = 21.5mg per 1kg rock

91.5% yield = 1kg rock
100% yield = 1kg rock x $^{100}/_{91.5}$
= 1.09kg rock

21.5mg found in 1.09kg rock
0.0215g found in 1.09kg rock
1.00g found in 1.09kg rock x $^{1.00}/_{0.0215}$
= 50.8kg rock</td>
</tr>
<tr>
<td>11a(i)</td>
<td>Lithium Aluminium Hydride or $LiAlH_4$</td>
<td>Lithium Aluminium Hydride is a reducing agent which would cause an aldehyde (methylpropanal) to be reduced to a primary alcohol (methylpropan-1-ol)</td>
</tr>
<tr>
<td>11a(ii)</td>
<td>Oxidation</td>
<td>Acidified dichromate solution containing $Cr_2O_7^{2-}$ ions is an oxidising agent.
This oxidising agent would oxidise an aldehyde (methylpropanal) to a carboxylic acid (methylpropanoic acid)</td>
</tr>
<tr>
<td>11a(iii)</td>
<td>Diagram showing:</td>
<td></td>
</tr>
<tr>
<td>11b(i)A</td>
<td>One circle drawn from:</td>
<td></td>
</tr>
</table>

<table>
<tr><td>11b(i)B</td><td>Diagram of
2,2-dimethylpropanoic acid</td><td></td></tr>
<tr><td>11b(ii)A</td><td></td><td>Examine closely original diagram to spot relationships between reactants and products

Make the changes to the reactants specified in the question and follow original
</td></tr>
<tr><td>11b(ii)B
PART I</td><td>Two different alpha hydrogen atoms/alpha hydrogen atoms present on different positions.
or
Butanone is asymmetrical.
or
Two different intermediates formed.
or
Intermediates formed have different stabilities.</td><td><table>
<tr><td>Carbon C_3 on left butanone molecule reacts with C_2 of right butanone molecule:</td><td>Carbon C_1 on left butanone molecule reacts with C_2 of right butanone molecule:</td></tr>
<tr><td></td><td></td></tr>
<tr><td>Product formed:
4-hydroxy-3,4-dimethylhexan-2-one</td><td>Product formed: (Compound Z)
5-hydroxy-5-methylheptan-3-one</td></tr>
</table></td></tr>
<tr><td>11b(ii)B
PART II</td><td>5-hydroxy-
5-methyl
heptan-3-one</td><td>$\underbrace{\text{5-hydroxy}}_{\text{-OH hydroxyl group on } C_5}\text{-}\underbrace{\text{5-methyl}}_{\text{-}CH_3\text{ methyl group on } C_5}\underbrace{\text{heptan}}_{\text{7 carbons on main chain}}\text{-}\underbrace{\text{3-one}}_{\text{C=O carbonyl group on } C_3}$</td></tr>
<tr><td>12a(i)</td><td>1st order</td><td><table>
<tr><td>Experiment</td><td>[$Na_2S_2O_3$]</td><td>[HCl]</td><td>Effect on Rate</td><td>Order of Reactant</td></tr>
<tr><td>❶+❷</td><td>x2</td><td>no change</td><td>x2</td><td>$[Na_2S_2O_3]^1$</td></tr>
<tr><td>❷+❸</td><td>no change</td><td>x2</td><td>no change</td><td>$[HCl]^0$</td></tr>
</table></td></tr>
<tr><td>12a(ii)</td><td></td><td>Zero Order Kinetics: rate of reaction is same at all concentrations of HCl</td></tr>
<tr><td>12a(iii)</td><td>Rate = $k[Na_2S_2O_3]$</td><td>rate = $k \times [Na_2S_2O_3]^1 \times [HCl]^0 = k[Na_2S_2O_3]$</td></tr>
</table>

12a(iv)	$0.131\ s^{-1}$	Using Experiment 1 (although experiments 2+3 can also be used) $k = \frac{rate}{[Na_2S_2O_3]} = \frac{6.55\times10^{-3}\ mol\ l^{-1}\ s^{-1}}{5.00\times10^{-2}\ mol\ l^{-1}} = 0.131\ s^{-1}$
12b(i)	monoclinic	
12b(ii)A		There is only one point on the graph where rhombic, monoclinic and vapour intersect. These conditions (~92°C and 10^{-5}atmospheres) are the only point where all three are in equilibrium with each other.
12b(ii)B	Rhombic, liquid and vapour	There is no junction of lines on the graph where rhombic, liquid and vapour meet